

Bill of materials and fault rates

For this example only the temperature correction factor is considered. Other factors are set to the value 1.
The average operating temperature, which has been determined with a temperature profile from the car manufacturer is assumed with 70°C
For the power components there is a temperature increase of 10°C assumed. Operating temperature is set to 80°C

Note: 1 FIT = 1e-9/h

Component name	Description	Value	Technology	Basis for fault rate	FIT value at ref. temperature	Reference temp. °C	Operat. temp. °C	Correction factor-temp, p(T)	Correction factor-voltage, p(U)	Correction factor-environment, p(E)	Correction factor-load, p(L)	Correction factor-failure criterion,	Correction factor-voltage, p(Q)	Correction factor-quality, p(D)	FIT Fault rate in use,
R1	Shunt Resistor		Wire Wound	SN29500	0.2	55	80	0.89	1	1	1	1	1	1	0.2
R2	Resistor		Metal film	SN29500	0.2	55	70	1.4	1	1	1	1	1	1	0.3
T1	Transistor	TO3	MOS-FET	SN29500	60	100	80	0.43	1	1	1	1	1	1	25.8
K1	Relay		Power	SN29500	30	70	70	1	1	1	1	1	1	1	30.0
D1	Diode		Standard	SN29500	1	40	70	3.7	1	1	1	1	1	1	3.7
IC1	Amplifier	AD8200	Dual Diff OP	SN29500	3	55	70	1.8	1	1	1	1	1	1	5.4
M	Motor		DC	Supplier	50			1	1	1	1	1	1	1	50.0